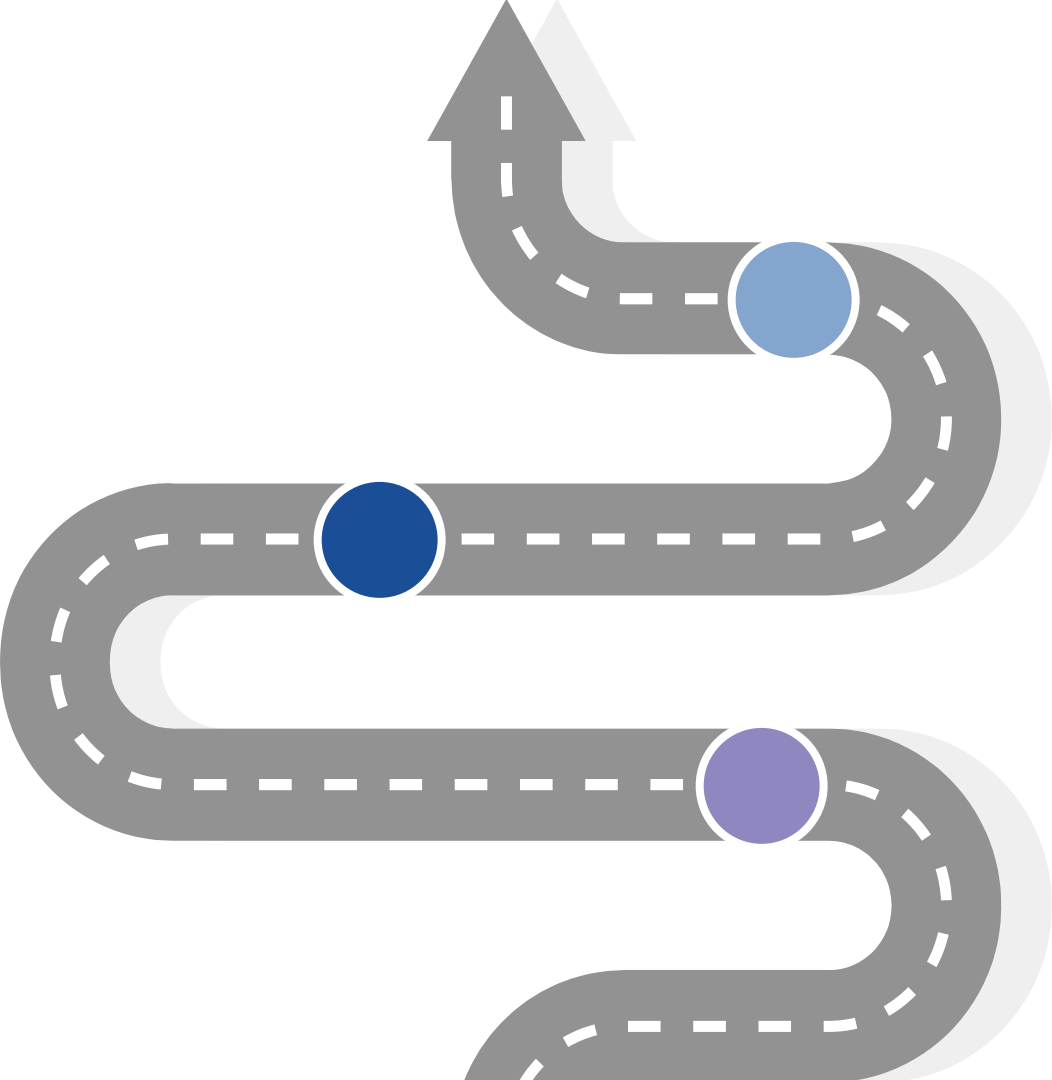




Does Daylight Savings Increase the Number of Fatal Vehicle Crashes?

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Original Paper Key Points

Smith, Austin C. "Spring Forward At Your Own Risk: Daylight Saving Time and Fatal Vehicle Crashes."

American Economic Journal. Applied Economics 2016, Vol. 8(2) 65-91. <https://www.jstor.org/stable/24739102>.

The issue of daylight savings time holds particular interest to us for a couple of reasons. One reason is that it is something that has been subject to political discourse about whether we should still change our clocks twice a year for daylight savings. Additionally, living in Boston shows how much daylight savings time can affect our days with the sun setting at its earliest at around 4:00 P.M. In his paper, Smith examines if there is any relationship between the number of vehicle crashes and deaths and daylight savings time. We believe that using the FARS and NHTSA data that is readily available will be able to help us replicate the data that shows that during daylight savings time there are more crashes in the evening when it is darker.

The original paper seeks to uncover if the change to Daylight Savings Time, especially the immediate week after, has any correlation with the number of motor vehicle fatalities. Smith hypothesizes that this is the case, due to both a shift in darkness hours when roads are busier, as well as, the impact of the one hour loss of sleep on the day of Daylight Savings.

Original Paper Method

Smith's original paper looks at data from 2002 to 2011, allowing five years on either side of the 2007 DST extension. Looking at national data for this 10 year period, Smith excludes Arizona and Indiana as parts of those states did not consistently practice DST. Furthermore, since the original day of DST is 23 hours, Smith adjusts the crash count by counting the 3-4 AM hour on the initial Sunday twice.

Smith uses **Regression Discontinuity (RD)** and **Fixed Effects (FE)** identification strategies. The RD estimates the effect if DST right at the spring transition, whereas this FE specification measures the average effect of DST over all dates that are sometimes DST and sometimes Standard Time during the spring. The RD model exploits the discrete change from Standard Time to DST where the regressions ran are centered at the transition date in each year. The coefficient of interest, Beta 1, is the aggregate effect of DST on vehicle data lies at the transition date.

$$\ln Fatal_{dy} = \beta_0 + \beta_1 DST_{dy} + f(DaysToTran) \\ + f(DST_{dy} \times DaysToTran_{dy}) + \varepsilon_{dy}.$$

The FE model Smith uses involve a 'DayofYear' dummy variable for each day of the year, as well as a 'DayofWeek' dummy variable to analyze the impact of DST on fatal vehicle crashes in the next immediate seven days following when DST starts each year.

$$\ln Fatal_{dy} = \beta_0 + \beta_1 SpDST_{dy} + \beta_2 FaDST_{dy} + DayofYear_d \\ + DayofWeek_{dy} + Year_y + \varepsilon_{dy}.$$

Data Source:

All of our data comes from FARS, which stands for Fatality Analysis Reporting System. This data comes from the National Highway Traffic Safety Association and contains data on a census of fatal traffic crashes within the 50 States, the District of Columbia, and Puerto Rico. In order to be included in FARS, the crash must have a motor vehicle traveling on a traffic way customarily open to the public, and must result in the death of a vehicle occupant or a non occupant within 30 days of the crash. We got our stata files directly from the National Bureau of Economic Research and for each year from 2002-2011, we compiled those years into an excel spreadsheet before exporting and working with the data within Stata.

Websites used:

- [Fatality Analysis Reporting System \(FARS\) | NHTSA](#)
- [Fatality Analysis Reporting System \(FARS\) | NBER](#)

Snippet of our Collapsed Data:

	day_week	year	date	dstdate	fatals	lnfatals	days	day_year	dstweek	detrend
1	3	2002	01jan2002	07apr2002	158	5.062595	-96	1	0	.3366331
2	4	2002	02jan2002	07apr2002	87	4.465988	-95	2	0	-.2870258
3	5	2002	03jan2002	07apr2002	95	4.553877	-94	3	0	-.226029
4	6	2002	04jan2002	07apr2002	110	4.70048	-93	4	0	-.1063975
5	7	2002	05jan2002	07apr2002	88	4.477337	-92	5	0	-.3565131
6	1	2002	06jan2002	07apr2002	107	4.672829	-91	6	0	.000811
7	2	2002	07jan2002	07apr2002	87	4.465988	-90	7	0	-.2330817
8	3	2002	08jan2002	07apr2002	85	4.442651	-89	8	0	-.2833105
9	4	2002	09jan2002	07apr2002	92	4.521789	-88	9	0	-.2311452
10	5	2002	10jan2002	07apr2002	123	4.812184	-87	10	0	.0322784
11	6	2002	11jan2002	07apr2002	109	4.691348	-86	11	0	-.1155299
12	7	2002	12jan2002	07apr2002	133	4.890349	-85	12	0	.0564989
13	1	2002	13jan2002	07apr2002	91	4.510859	-84	13	0	-.1611582
14	2	2002	14jan2002	07apr2002	94	4.543295	-83	14	0	-.1556948
15	3	2002	15jan2002	07apr2002	93	4.532599	-82	15	0	-.1933624
16	4	2002	16jan2002	07apr2002	108	4.682131	-81	16	0	-.0780826
17	5	2002	17jan2002	07apr2002	94	4.543295	-80	17	0	-.236611
18	6	2002	18jan2002	07apr2002	109	4.691348	-79	18	0	-.1155299
19	7	2002	19jan2002	07apr2002	136	4.912655	-78	19	0	.0788049
20	1	2002	20jan2002	07apr2002	94	4.543295	-77	20	0	-.1287228
21	2	2002	21jan2002	07apr2002	81	4.394449	-76	21	0	-.3045495
22	3	2002	22jan2002	07apr2002	92	4.521789	-75	22	0	-.2041732
23	4	2002	23jan2002	07apr2002	75	4.317488	-74	23	0	-.4354457
24	5	2002	24jan2002	07apr2002	93	4.532599	-73	24	0	-.2473865
25	6	2002	25jan2002	07apr2002	135	4.905275	-72	25	0	.0983969
26	7	2002	26jan2002	07apr2002	134	4.89784	-71	26	0	.06399
27	1	2002	27jan2002	07apr2002	111	4.70953	-70	27	0	.0375127
28	2	2002	28jan2002	07apr2002	98	4.584968	-69	28	0	-.1140221
29	3	2002	29jan2002	07apr2002	74	4.304065	-68	29	0	-.4218966
30	4	2002	30jan2002	07apr2002	84	4.430817	-67	30	0	-.3221172
31	5	2002	31jan2002	07apr2002	92	4.521789	-66	31	0	-.2581173
32	6	2002	01feb2002	07apr2002	113	4.727388	-65	32	0	-.0794901
33	7	2002	02feb2002	07apr2002	111	4.70953	-64	33	0	-.1243197
34	1	2002	03feb2002	07apr2002	121	4.795701	-63	34	0	.133773

. summ

Variable	Obs	Mean	Std. dev.	Min	Max
day_week	3,657	4.008203	2.007082	1	9
year	3,657	2006.497	2.872387	2002	2011
date	3,652	17166.5	1054.386	15341	18992
dstdate	3,657	17064.76	1038.373	15437	18699
fatals	3,657	102.3459	26.66299	1	191
Infatals	3,657	4.591524	.2952132	0	5.252274
days	3,652	100.7004	106.1615	-96	298
day_year	3,652	183.1002	105.4386	1	366
dstweek	3,657	.0191414	.1370406	0	1
detrend	3,657	-1.31e-10	.2711845	-4.887794	.7585776

. summ if abs(days <= 48.5)

Variable	Obs	Mean	Std. dev.	Min	Max
day_week	1,304	4.032209	2.000891	1	7
year	1,304	2006.262	2.878494	2002	2011
date	1,304	16962.96	1046.678	15341	18747
dstdate	1,304	16980.24	1040.59	15437	18699
fatals	1,304	94.6388	24.46676	35	176
Infatals	1,304	4.516958	.2585788	3.555348	5.170484
days	1,304	-17.28604	38.65355	-96	48
day_year	1,304	66.28604	38.65355	1	145
dstweek	1,304	.053681	.2254735	0	1
detrend	1,304	-.083645	.2199156	-.8740965	.5049109
_est_model~E	1,304	1	0	1	1
_est_model~1	1,304	.4217791	.4940331	0	1
_est_model~2	1,304	.743865	.4366647	0	1
_est_model~3	1,304	.5138037	.5000012	0	1

Summary Statistics

. describe

Contains data from L:\0414 COLLAPSED DATA SET.dta
Observations: 3,657
Variables: 14 14 Apr 2024 14:56

Variable name	Storage type	Display format	Value label	Variable label
day_week	byte	%10.0g		day_week
year	int	%10.0g		year
date	float	%td		
dstdate	float	%td		
fatals	double	%10.0g		(sum) fatalities
Infatals	float	%9.0g		
days	float	%9.0g		
day_year	float	%9.0g		
dstweek	float	%9.0g		
detrend	float	%9.0g		Residuals

Summary Statistics, What Went Right

The paper does not include specific summary statistics, so it is impossible to say with absolute certainty how accurately our summary statistics line up with the paper. However, we used the same FARS data sets as the paper and included the same variables as the paper, dropping the same variables and data entries as the paper did. Our data set was built following the same steps as the paper did in building the data set. Additionally, our number of total observations line up with the number of observations in the paper.

Summary Statistics, What Went Wrong

When we had to run our regressions, we used a slightly different date time frame than the paper in order to have the regressions line up correctly. This could imply differences in the summary statistics in our larger data set as a whole.

```
. summ if abs(days <= 48.5)
```

Variable	Obs	Mean	Std. dev.	Min	Max
day_week	1,304	4.032209	2.000891	1	7
year	1,304	2006.262	2.878494	2002	2011
date	1,304	16962.96	1046.678	15341	18747
dstdate	1,304	16980.24	1040.59	15437	18699
fatals	1,304	94.6388	24.46676	35	176
lnfatals	1,304	4.516958	.2585788	3.555348	5.170484
days	1,304	-17.28604	38.65355	-96	48
day_year	1,304	66.28604	38.65355	1	145
dstweek	1,304	.053681	.2254735	0	1
detrend	1,304	-.083645	.2199156	-.8740965	.5049109
_est_model~E	1,304	1	0	1	1
_est_model~1	1,304	.4217791	.4940331	0	1
_est_model~2	1,304	.743865	.4366647	0	1
_est_model~3	1,304	.5138037	.5000012	0	1

Regression Results - RD Method

```
. esttab model_table1_1 model_table1_2 model_table1_3, r2 ar2 se
```

	(1) detrnd	(2) detrnd	(3) detrnd
days	0.00240*** (0.000594)	0.00175*** (0.000256)	0.00164*** (0.000440)
dstweek	0.0609* (0.0283)	0.0530 (0.0277)	0.0610* (0.0278)
_cons	-0.0854*** (0.0101)	-0.0756*** (0.00744)	-0.0832*** (0.00897)
N	550	970	670
R-sq	0.039	0.050	0.029
adj. R-sq	0.036	0.048	0.026

Standard errors in parentheses
 * p<0.05, ** p<0.01, *** p<0.001

Our Results

TABLE 8—RD ESTIMATES OF THE IMPACT OF ENTERING DST ON FATAL CRASHES ACROSS TIME (*Spring*)

	2002–2011			1976–2001		
	(1)	(2)	(3)	(4)	(5)	(6)
DST	0.0649*** (0.0231)	0.0499*** (0.0176)	0.0626*** (0.0215)	0.0398** (0.0176)	0.00264 (0.0116)	0.0420** (0.0180)
Bandwidth selector	CCT	IK	CV	CCT	IK	CV
Observations	550	966	670	1,014	2,155	962

Notes: Dependent variable is log fatal crashes demeaned by day-of-week and year. All specifications use a first order polynomial and a uniform kernel. DST is the estimate of the discontinuity in fatal crashes that occurs immediately following the spring transition into DST. CCT refers to the bandwidth selector of Calonico, Cattaneo, and Titiunik (2014a); IK is Imbens and Kalyanaraman (2012); CV is the cross-validation method of Ludwig and Miller (2007). Robust standard errors are in parentheses.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

Source: Authors' calculations

Smith's Results

Regression Results - FE Method

Our Results

	(1) Infatals
days	-0.00758*** (-36.99)
dstweek	0.0452* (2.01)
1.day_week	0 (.)
2.day_week	-0.285*** (-29.62)
3.day_week	-0.328*** (-34.46)
4.day_week	-0.296*** (-30.92)
5.day_week	-0.249*** (-25.99)
6.day_week	-0.0397*** (-4.17)
7.day_week	0.125*** (12.94)
_cons	5.513*** (253.90)
N	3652
R-sq	0.682
adj. R-sq	0.646
RMSE	

t statistics in parentheses
 * p<0.05, ** p<0.01, *** p<0.001

TABLE 5—FE ESTIMATES OF THE IMPACT OF DST ON FATAL CRASHES—DECOMPOSING SPRING DST

	All hours				Least light impacted	Morning	Evening
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Spring DST	0.0319* (0.0165)	0.0307* (0.0165)					
First six days of DST			0.0565** (0.0231)	0.0559** (0.0230)	0.0574** (0.0272)	0.205*** (0.0514)	-0.0265 (0.0453)
Next eight days of DST			0.0254 (0.0201)	0.0240 (0.0201)	0.0289 (0.0234)	0.130** (0.0603)	-0.0812* (0.0450)
Remainder of spring DST			0.0142 (0.0197)	0.0123 (0.0197)	0.00907 (0.0230)	0.126** (0.0553)	-0.0588 (0.0429)
Fall DST	0.0228 (0.0249)	0.0221 (0.0247)	0.0218 (0.0250)	0.0211 (0.0248)	0.0446 (0.0303)	0.259*** (0.0709)	-0.159*** (0.0482)
ln(gas price)		-0.0457* (0.0246)		-0.0469* (0.0247)	-0.0449 (0.0289)	-0.101* (0.0557)	-0.0307 (0.0484)
Observations	3,341	3,341	3,341	3,341	3,341	3,341	3,341
Adjusted R ²	0.734	0.735	0.735	0.735	0.753	0.184	0.319

Notes: Dependent variable is the log fatal crashes; all specifications use day-of-year, day-of-week, and year dummies. Remainder of spring DST is an indicator variable equal to one if the day occurs after the first two weeks of DST and before July 1st. Fall DST is an indicator variable equal to one if the day falls under DST and occurs after June 30th. "Morning" is defined as +/- two hours from the average sunrise time in that location around both transitions; "Evening" is defined as +/- two hours from the average sunset time in that location around both transitions; least light impacted are the remaining hours. Robust standard errors are in parentheses.

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

* Significant at the 10 percent level.

Source: Authors' calculations

Smith's Results

Regression Results - additional *areg* and *FE*

```
. areg lnfatals days dstweek i.day_week, absorb(day_year)
```

Linear regression, absorbing indicators
Absorbed variable: day_year

Number of obs = 3,652
No. of categories = 366
F(8, 3278) = 702.55
Prob > F = 0.0000
R-squared = 0.6822
Adj R-squared = 0.6461
Root MSE = 0.1531

Our Results

lnfatals	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
days	-.0075788	.0002049	-36.99	0.000	-.0079804	-.0071771
dstweek	.0451638	.0225187	2.01	0.045	.0010117	.089316
day_week						
2	-.2853121	.0096311	-29.62	0.000	-.3041957	-.2664285
3	-.3283646	.0095279	-34.46	0.000	-.3470458	-.3096834
4	-.2962274	.0095796	-30.92	0.000	-.31501	-.2774448
5	-.248956	.0095777	-25.99	0.000	-.2677349	-.2301771
6	-.039721	.0095297	-4.17	0.000	-.0584058	-.0210361
7	.1245952	.0096268	12.94	0.000	.10572	.1434703
_cons	5.512579	.0217117	253.90	0.000	5.470009	5.555149

F test of absorbed indicators: F(365, 3278) = 6.935

Prob > F = 0.000

TABLE 5—FE ESTIMATES OF

	(1)
Spring DST	0.0319* (0.0165)

Smith's Results

Regression, What Went Right

RD Models:

- Our Regression Model 1-3 replicates Smith's Table 8 Model 1-3. Our number of observations for Model 1 and Model 3 are identical. This is because of our regression window of 27.5 and 33.5 days, as shown in the code to the right. Our Model 2 is also only four observations away from Smith's model. This success is likely due to the day windows we used, as opposed to sticking with 45 days as he writes in his paper.
- Our *dstweek* coefficients are relatively close to Smith's, all being at most 0.004 above or below Smith's coefficients. This success is likely due to our close amount of observations to Smith's model. This aids in a consistent sample size as we also succeed in other summary statistic steps.
- Smith and our standard errors (shown in the parentheses below each coefficient) are relatively close to each other, with the exception of Model 2.

FE Model:

- Our Fixed Effects Model looks at our *dstweek* coefficient and Smith's Table 5 Regression 1 *Spring DST* coefficient.
- Our FE regression *dstweek* coefficient is statistically significant at the 95% level as is Smith's Spring DST coefficient. This success is likely due to the fact that a 95% confidence interval is slightly broader and therefore captures more values. Furthermore, our data steps for the week entering DST aligned with Smith's.

```
reg detrend days dstweek if abs(days) <= 27.5
eststo model_table1_1

reg detrend days dstweek if abs(days) <= 48.5
eststo model_table1_2

reg detrend days dstweek if abs(days) <= 33.5
eststo model_table1_3

esttab model_table1_1 model_table1_2 model_table1_3, r2 ar2
```

Regression, What Went Wrong

RD Models:

- Our regression *dstweek* coefficient is statistically only at the 95% level, where as Smith's is significant at the 99%. This may be due to our lack of Bandwidth selector, as well as, our method to achieve 550, 970, or 670 observations (using, for example, a 48.5 day window, opposed to a 45 day window to achieve an accurate number of observations).
- The standard error of Model 2 in our Regression is further away from Smith's, likely due to the number of observations (970 vs 966). The Standard Error is 0.0101 away from Smith's coefficient (0.0277 vs 0.0176).
- Our RD Adjusted R^2 is low (0.036). This may be due to the variables we included or did not include in our model. While we did regress *dstweek* and *days* on *detrend*, there of course are explanatory variables left out of the model. We were not given Smith's exact regression so this may have caused the discrepancy in our regression.

FE Model:

- Our FE Adjusted R^2 is smaller than Smith's ($0.646 < 0.734$). This is likely due to the difference in observations and variables ran in the regression, as described below.
- Our number of observations is greater than Smith's ($3,652 > 3,341$) which may be due to the window of dates we ran our fixed effects model with. Our code (`reg fatalities i.day_week`), as well as the areg code (`areg ln(fatalities) days dstweek i.dow if abs(days) <= 45, absorb(doy)`) may have used a different observations window. Furthermore, for the fixed effects model, we are not definitively sure if he ran the model with other variables than *dstweek* or *days* as his Table 5 displays other results, i.e. for Fall DST.
- Our *dstweek* coefficient is also 0.0133 off from Smith's coefficient and, as seen in our *areg* function in Slide 10, our Standard Error for *dstweek* is also more than 0.028 off from Smith's standard error.

Conclusion

To conclude our replication, the main question that we wanted to answer was: Does Daylight Savings Increase the Number of Fatal Vehicle Crashes? This paper appealed to us as this question itself was interesting, with daylight savings being a nationwide practiced policy, and the impact that it has on car crashes is an impact that one would not think about right away. Through our process of collecting data from FARS and merging the different years from 2002 to 2011, we focused on certain windows of 27.5 days, 48.3 days, and 33.5 days, while Smith focused on a window about 45 days. Smith concluded that the estimated impact of DST on fatal crashes is about an increased risk of about 5-6.5 percent. Through our regressions, we found a similar conclusion that DST's estimated impact on fatal car crashes is about 5.3-6.1 percent in the spring-time window. Also, our fixed effects regression model is statistically significant at the 99% level, and our fixed effects regression dstweek coefficient is statistically significant at the 95% level, which correlates to Smith's Spring DST coefficient. Smith went further into the effect of this sleep deprivation effect, which is something that we also believe could be another interesting aspect of Daylight Savings that has yet to be discovered further about and believe that our findings could contribute to the supporting evidence in the effects of Daylight Saving on our daily lives.

